MCO-1D

Commissioning Manual

Pre-build, pre-flight, and first-flight procedures

v25 · September 2026

This version adds pre-build resolution items from the adversarial physics audit, ODrive regen configuration, and the first-flight envelope characterisation procedure (H-PERF-6).

1. Pre-Build Resolution Items

These must be resolved before the build is finalised. The mass budget issue is the most urgent.

CRITICAL **Mass budget resolution**

Confirm final airframe build mass against MTOW_KG=3.448 kg. Current estimate M_AIRFRAME_KG=4.1 kg exceeds MTOW. Options: (a) achieve lighter build, (b) increase wing geometry, (c) increase max frequency ceiling, (d) accept forward-flight-only mode and rely on H-SAR-FF forward-flight SAR family.

HIGH **Structural load testing**

Static load test TPU winglet hinges before any stoop or high-speed manoeuvres. Test across temperature range (0°C to 40°C). Verify STOOP_MAX_WING_LOAD_G = 4.0 or update. TPU Shore 85A assumed.

MEDIUM **ODrive regen enable**

odrv0.config.dc_bus_max_negative_current = -5.0 on all 8 ODrives. Then odrv0.save_configuration(). Verify with regen_status after first powered descent. Battery must be below 95% SOC for regen to activate.

MEDIUM **Servo programming**

All 16 servos (8 pitch + 8 sweep): program to 170° max travel via AGF-SPV3 programmer before installation. AGFRC IB53BHU units.

MEDIUM **Encoder magnet check**

mco_odrive_setup.py --check-magnet — verify all 8 encoder magnets installed and returning valid signals before wing installation.

MEDIUM **Wing commander constants**

wc_set_fuselage_width and wc_set_tpu_start must be set from as-built measurements before TC-01. Pre-flight check will not pass until these are populated.

2. Pre-Flight Checklist Additions

Add to the standard pre-flight checklist before first flight:

- ■ ALC modalities all healthy: alc_status → check modalities section
- ■ Regen configured: regen_config_advice printed and executed on all ODrives
- ■ H-PERF-6 is first in queue: disc_registry → verify priority 0 at top
- ■ HOVER_CAPABLE = None: envelope_status → confirm unmeasured

■ Wear baseline: wear_status → baseline_set = False (expected on first flight)

■ ALC phase: alc_set_flight_phase pre_flight → alc_preflight_brief

■ Battery < 95% SOC: required for regen to activate at full rate

3. First-Flight Sequence (H-PERF-6)

H-PERF-6 runs automatically as the first hypothesis (priority 0). The pilot does not need to command it. The procedure below is what the pilot should attempt during the first flight to populate the envelope.

Phase 1 — Launch and cruise

Establish stable cruise at 9 m/s, 20m AGL. Confirm all systems nominal. Allow H-PERF-6 to collect baseline data.

Phase 2 — Deceleration survey

Reduce lever to trim detent. Slow in 0.5 m/s steps. Hold each speed 10 seconds. Watch VSI. Stop if VSI < –0.5 m/s.

Phase 3 — Hover attempt (if V_min < 3 m/s)

Slow toward 0 m/s with lever at trim detent. If V < 1.0 m/s maintained 10s with level flight → hover capable. Record endurance.

Phase 4 — Landing

Use H-PREC-2 profile. Approach at V_min. Flare at 2.5m. Aim for <2 m/s horizontal at touchdown.

Phase 5 — Post-flight debrief

GS: alc_debrief. Review envelope section. Then: set_hover_capable true/false. Persist: persist_save_all.

4. Wear Baseline Establishment

The mechanical wear modality establishes baseline signatures after 3 flights. During this period it collects data silently. No action required. After flight 3, wear_status will show baseline_set = True and per-wing wear indices will begin tracking.

After baseline is set: check wear_status after every 5 flights. Advisory thresholds: hinge friction 12%/22%, stiffness 10%/20%, winding resistance 8%/15% above baseline. Service intervals: 10,000 strokes minor, 50,000 strokes full.